# NovaCloud Enterprise Architecture & High-Availability SLA (2026)

*Master Infrastructure Agreement, Availability Targets, and Operational Protocol*

## 1. Service Level Objectives (SLO) & Uptime Commitments

NovaCloud guarantees enterprise-tier infrastructure availability and API latency standards across multi-region production clusters:

- **Monthly Uptime Commitment:** NovaCloud commits to a 99.99% monthly availability percentage across all core API endpoints, database clusters, and container runtime nodes.
- **API Latency Standard:** 99th percentile (p99) API response latency must remain strictly below 25 milliseconds under standard operating load of up to 50,000 requests per second.
- **Scheduled Maintenance Windows:** Maintenance operations are restricted to Sundays between 02:00 AM and 04:00 AM UTC with a mandatory 72-hour advance advisory notice.

## 2. Disaster Recovery, Replication, and Resiliency

Our distributed architecture provides automated failover and data resilience across geographically isolated availability zones:

- **Cross-Region Replication:** Database mutations are replicated asynchronously across primary and secondary regions using distributed consensus with a replication lag below 300ms.
- **Recovery Point Objective (RPO):** NovaCloud guarantees an RPO of less than 5 minutes for stateful transactional data.
- **Recovery Time Objective (RTO):** Guaranteed automated cluster failover RTO of less than 15 minutes during total regional failure.

## 3. Cryptographic Controls and Data Protection at Rest and Transit

Enterprise tenant data is insulated with defense-in-depth cryptographic controls:

- **Encryption at Rest:** All block storage, object storage buckets, and relational table snapshots are encrypted using AES-256 GCM.
- **Customer Managed Encryption Keys (CMEK):** Customers may integrate external Cloud KMS with mandatory cryptographic key rotation every 180 days.
- **Transport Layer Security:** All incoming and internal inter-service network ingress mandates TLS version 1.3 with forward secrecy ciphers.

## 4. Incident Severity Classification & SLA Financial Credits

In the event of an unscheduled degradation, NovaCloud applies structured incident escalation and financial penalty offsets:

- **Severity 1 (Critical Outage):** Total unavailability of core API or database service. Dedicated engineering incident commander responds within 15 minutes 24/7/365.
- **Severity 2 (High Degradation):** Substantial latency elevation or loss of redundancy. Response SLA is within 45 minutes.
- **SLA Remedy Credits:** If monthly uptime falls between 99.0% and 99.98%, customer receives a 10% credit. If uptime drops below 99.0%, customer receives a 25% credit applied to next billing invoice.